Comparison of Generative Models on the ArtBench-10 Dataset

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Abstract. This paper presents the implementation and comparative evaluation of three generative image model families — a Variational Autoencoder (VAE), a Deep Convolutional Generative Adversarial Network (DCGAN), and a Denoising Diffusion Probabilistic Model (DDPM) — trained on the ArtBench-10 dataset (32×32 pixels, 10 artistic styles). Models were first developed on a 20% data subset and subsequently evaluated on the full 50,000-image training set. Quantitative evaluation uses Fréchet Inception Distance (FID) and Kernel Inception Distance (KID), each reported as mean $\pm$ standard deviation over 10 random seeds with 5,000 generated samples. The DCGAN achieved the best results (FID = 30.46 ± 0.57; KID = 0.0032 ± 0.0001), followed by the DDPM (FID = 37.91 ± 0.37; KID = 0.0062 ± 0.0002) and the VAE (FID = 140.37 ± 0.43; KID = 0.0314 ± 0.0004). We analyse the trade-offs in sample quality, diversity, training stability, and computational cost.

Keywords: Generative Models · VAE · DCGAN · DDPM · ArtBench-10 · FID · KID

1 Introduction

Generative modelling of images has become a central research area, with applications ranging from data augmentation to creative content synthesis. Three paradigms have emerged as dominant: Variational Autoencoders (VAE) [6], Generative Adversarial Networks [8], and Diffusion Models [3], each offering distinct trade-offs between sample fidelity, diversity, training stability, and inference cost.

Evaluating these models in the artistic domain is particularly challenging because artworks exhibit complex textures, high style variability, and require both visual realism and stylistic coherence. The ArtBench-10 dataset [7] was designed precisely to benchmark generative models in this setting, providing a curated, class-balanced collection of artwork images across 10 historical styles.

This work addresses the task of learning generative models from artwork images and producing new plausible samples that match the data distribution. We build a complete generative pipeline — covering data loading, preprocessing, training, sampling, and evaluation — and compare the three model families under a controlled, reproducible protocol. All models were initially developed on a 20% data subset for rapid iteration and then trained on the full dataset for final evaluation.

The remainder of this paper is organised as follows: Section 2 describes the problem and dataset; Section 3 details the three model architectures; Section 4 describes the experimental setup; Section 5 presents and analyses the results; and Section 6 concludes the paper.

2 Problem Description

The goal of this project is image generation, not classification. Given a dataset of artwork images, a generative model must learn the underlying data distribution and produce new samples that are visually plausible and statistically consistent with the training data.

Dataset: ArtBench-10

ArtBench-10 [7] is a curated benchmark dataset for generative modelling in the visual arts domain. It contains paintings grouped by artistic style, specifically designed to evaluate models such as VAEs, GANs, and diffusion models on structured artistic data. Its main characteristics are:

- **Classes:** 10 artistic styles — *Art Nouveau*, *Baroque*, *Expressionism*, *Impressionism*, *Post-Impressionism*, *Realism*, *Renaissance*, *Romanticism*, *Surrealism*, and *Ukiyo-e*.
- **Size:** 50,000 training images and 10,000 test images (5,000/1,000 per class — balanced).
- **Resolution:** 32×32 RGB.

This setting is challenging because models must simultaneously balance visual fidelity, sample diversity, and style consistency. Unlike discriminative tasks, quality must be assessed both qualitatively (sample inspection) and quantitatively (distribution-level metrics).

Development Protocol

Following the prescribed protocol, models were initially trained and tuned on a provided **20% data subset** (10,000 images) to allow fast iteration over architectures and hyperparameters. All three model families were then retrained on the **full training set** (50,000 images) for the final evaluation, going beyond the minimum requirement of retraining only the best-performing model.

3 Methodology

3.1 Variational Autoencoder (VAE)

The VAE [6] learns a latent representation of the data by maximising the Evidence Lower BOund (ELBO):

$$\mathcal{L}(\theta, \phi;\, x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - D_{\mathrm{KL}}(q_\phi(z|x) \,\|\, \mathcal{N}(0, I))\,, \tag{1}$$

where the encoder $q_\phi(z|x) = \mathcal{N}(\mu_\phi, \sigma_\phi^2 I)$ maps inputs to a Gaussian posterior and the decoder $p_\theta(x|z)$ reconstructs the input. Sampling is enabled by the reparameterisation trick $z = \mu + \sigma \odot \varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$.

We use a β-VAE formulation with $\beta = 0.5$ to weight reconstruction more heavily than regularisation: $\mathcal{L}_\beta = \mathcal{L}_{\text{recon}} - \beta\, D_{\text{KL}}$.

Architecture. *Encoder:* three strided conv. blocks ($3 \to 64 \to 128 \to 256$ channels, stride 2, BatchNorm, LeakyReLU 0.2), then linear heads to $\mu, \log\sigma^2 \in \mathbb{R}^{128}$. *Decoder:* three transposed-conv. blocks ($256 \to 128 \to 64 \to 3$, BatchNorm, ReLU, Sigmoid).

3.2 Deep Convolutional GAN (DCGAN)

GANs [8] train a generator G and discriminator D via a minimax objective:

$$\min_G \max_D \; \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z\sim p_z}[\log(1 - D(G(z)))]\,. \tag{2}$$

In practice the generator maximises $\mathbb{E}_z[\log D(G(z))]$ to avoid gradient saturation. DCGAN [8] stabilises training via strided convolutions (no pooling), BatchNorm, and LeakyReLU in the discriminator.

Architecture. The generator maps $z \in \mathbb{R}^{100}$ through four transposed-convolutional blocks ($512{\times}4{\times}4 \to 256 \to 128 \to 3{\times}32{\times}32$, Tanh output). The discriminator applies four strided convolutions ($3 \to 64 \to 128 \to 256 \to 1$, Sigmoid output, LeakyReLU 0.2). Weights are initialised as $\mathcal{N}(0, 0.02)$. Label smoothing sets real labels to 0.9 to prevent discriminator overconfidence.

3.3 Denoising Diffusion Probabilistic Model (DDPM)

DDPMs [3] define a Markov forward process that adds Gaussian noise over T steps:

$$q(x_t|x_0) = \mathcal{N}\big(\sqrt{\bar\alpha_t}\, x_0,\ (1-\bar\alpha_t)I\big)\,, \quad \bar\alpha_t = \textstyle\prod_{s=1}^{t}(1-\beta_s), \tag{3}$$

and learn a neural network ϵ_θ to reverse this process via:

$$\mathcal{L} = \mathbb{E}_{t,x_0,\epsilon}\Big[\big\|\epsilon - \epsilon_\theta\big(\sqrt{\bar\alpha_t}\, x_0 + \sqrt{1{-}\bar\alpha_t}\,\epsilon,\ t\big)\big\|^2\Big]\,. \tag{4}$$

Recent work [1] has shown that diffusion models can surpass GANs in image quality at high resolutions. For efficient inference we apply DDIM sampling [1] with 100 steps instead of the 1,000 training steps, reducing generation time by a factor of 10.

Architecture. A U-Net with 29.60 M parameters operates at three resolution levels ($32{\to}16{\to}8$ pixels). Each level contains two ResBlocks (GroupNorm, SiLU, Conv2d $3{\times}3$, Dropout 0.1, residual connection). 4-head self-attention is applied at the $16{\times}16$ and $8{\times}8$ levels. Timestep t is encoded via a sinusoidal positional embedding followed by a two-layer MLP and injected into every ResBlock via a linear projection. An Exponential Moving Average (EMA, decay = 0.9995) of the weights is maintained and used exclusively during sampling.

Table 1. Training hyperparameters for each model (full dataset).

Hyperparameter	VAE	DCGAN	DDPM
Epochs	100	200	500
Batch size	64	64	128
Optimiser	Adam	Adam	AdamW
Learning rate	10^{-3}	2×10^{-4}	2×10^{-4}
Adam β_1	0.9	0.5	0.9
Latent / noise dim	128	100	—
Noise steps T	—	—	1000
LR schedule	None	None	Warmup + cosine
EMA decay	—	—	0.9995
Total grad. steps	∼78k	∼156k	∼195k

Table 2. Model size and inference cost.

Model	Parameters	Grad. steps	Inference passes
VAE	∼2.9 M	∼78k	1
DCGAN	∼4.1 M (G+D)	∼156k	1
DDPM	29.60 M	∼195k	100 (DDIM)

4 Experimental Setup

4.1 Data Preprocessing

All images are resized and centre-cropped to 32×32 pixels. The VAE normalises pixel values to $[0, 1]$ (compatible with its Sigmoid output); the DCGAN and DDPM normalise to $[-1, 1]$ (Tanh and standard normalisation, respectively). The DDPM additionally applies random cropping with a 4-pixel margin, random horizontal flipping ($p = 0.5$), and mild colour jitter (brightness/contrast/saturation $\pm5\%$, hue $\pm2\%$) for data augmentation.

4.2 Training Configuration

Table 1 summarises the main hyperparameters. All experiments used a fixed random seed (42) for reproducibility. Training was performed on a single NVIDIA GPU with CUDA support.

4.3 Computational Resources

Table 2 summarises model sizes and training costs. The VAE and DCGAN are lightweight: a single forward pass suffices to generate one sample. The DDPM is $7\times$ larger and requires 100 denoising steps per image even with DDIM acceleration.

Fig. 1. 64 generated samples (full dataset, 32×32 px). Left: VAE. Centre: DCGAN. Right: DDPM.

4.4 Evaluation Protocol

The evaluation protocol follows the assignment specification exactly:

- **Samples:** 5,000 images generated by each model and 5,000 real images drawn from the test set.
- **FID:** computed on the full 5,000-sample sets using InceptionV3 features (2,048-dimensional, fc replaced by `Identity`).
- **KID:** computed as the mean MMD^2 over 50 random subsets of size 100 drawn from the generated and real feature sets; both the mean and standard deviation are reported per run.
- **Statistical robustness:** the full protocol is repeated for 10 different random seeds (42–51). Final results are reported as mean $\pm$ standard deviation of FID and mean $\pm$ standard deviation of KID across the 10 runs.
- **Fairness:** preprocessing, sample count, and evaluation code path are identical for all models. DDIM sampling (100 steps) is used for the DDPM.

5 Results and Analysis

5.1 Qualitative Analysis

Figure 1 shows 64 samples generated by each model after training on the full dataset.

The VAE produces noticeably blurry images, a direct consequence of the pixel-wise BCE reconstruction loss which averages over plausible outputs. The DCGAN generates sharper, more detailed images but shows some repetition of motifs, suggesting mild mode dropping. The DDPM produces the most varied outputs, covering a wider range of styles and colour palettes, reflecting the richer probabilistic formulation of the reverse diffusion process.

The VAE training curves (Figure 2) confirm stable convergence: the total loss decreased from $\approx$1840 to $\approx$1780 over 100 epochs, with the KL term stabilising at $\approx$65, indicating a well-formed latent space.

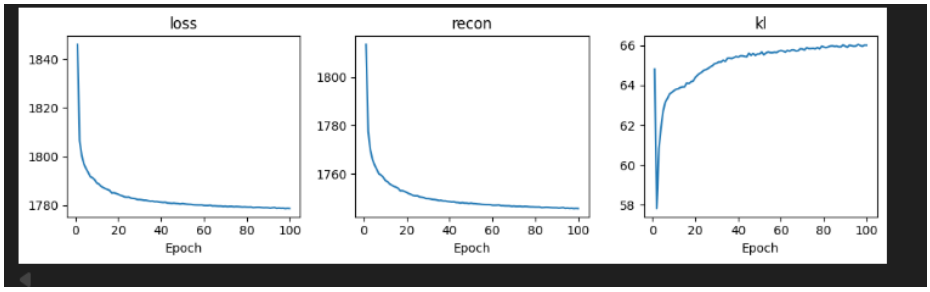

Fig. 2. VAE training curves: total loss (left), reconstruction BCE (centre), and KL divergence (right) over 100 epochs.

Table 3. FID and KID (mean ± std over 10 seeds, full dataset). Bold indicates best value.

Model	**FID** (↓)	**KID** (↓)
VAE	140.37 ± 0.43	0.0314 ± 0.0004
DCGAN	$\mathbf{30.46 \pm 0.57}$	$\mathbf{0.0032 \pm 0.0001}$
DDPM	37.91 ± 0.37	0.0062 ± 0.0002

5.2 Quantitative Analysis

Table 3 reports the main quantitative results.

Figure 3 visualises these results; Table 4 provides the full per-seed breakdown.

Table 4. Per-seed FID and KID for all models (full dataset, seeds 42–51).

	FID			**KID**		
Seed	VAE	DCGAN	DDPM	VAE	DCGAN	DDPM
42	140.66	31.10	37.91	0.0315	0.0032	0.0061
43	140.34	29.56	38.35	0.0315	0.0033	0.0065
44	139.59	29.91	37.63	0.0311	0.0031	0.0063
45	140.29	30.29	37.88	0.0315	0.0032	0.0063
46	140.34	30.23	38.05	0.0306	0.0032	0.0063
47	141.28	31.35	37.39	0.0321	0.0033	0.0059
48	140.76	30.87	38.29	0.0315	0.0034	0.0063
49	140.07	30.70	37.20	0.0314	0.0032	0.0060
50	140.26	29.73	38.04	0.0319	0.0030	0.0062
51	140.08	30.82	38.33	0.0311	0.0032	0.0062
Mean	140.37	30.46	37.91	0.0314	0.0032	0.0062
Std	0.43	0.57	0.37	0.0004	0.0001	0.0002

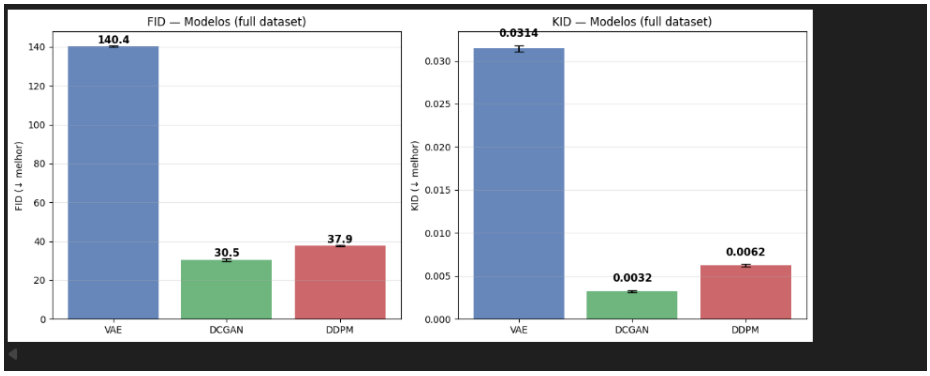

Fig. 3. FID (left) and KID (right) for the three models. Error bars represent one standard deviation over 10 seeds.

VAE. The high FID (140.37) reflects the known limitation of pixel-wise reconstruction losses: minimising the average reconstruction error encourages blurry outputs, since the mean of multiple plausible sharp images is itself blurry. The regularised latent space supports smooth interpolation but does not enforce perceptual sharpness.

DCGAN. The adversarial objective directly minimises the perceptual distance between real and generated distributions, explaining the strong FID (30.46) and KID (0.0032). At 32×32 resolution the discriminator's capacity is sufficient to model the full feature space. The slightly larger FID variance (± 0.57) reflects the inherent stochasticity of the GAN training dynamic. Mode dropping is visible qualitatively but not captured by FID/KID.

DDPM. The DDPM ranks between DCGAN and VAE (FID 37.91). Its lower FID variance (± 0.37) indicates more stable generation. The DDIM approximation (100 vs. 1,000 steps) likely accounts for part of the gap with DCGAN, as fewer denoising steps may not fully capture the learned distribution. At higher resolutions diffusion models typically surpass GANs [1]; at 32×32 the DCGAN's simpler objective is sufficient to achieve better distribution matching.

The very low standard deviations across all models and metrics confirm that 10 seeds with 5,000 samples provide statistically robust estimates, in line with the evaluation protocol.

5.3 Development Subset vs. Full Dataset

Table 5. FID and KID on the 20% development subset (seeds 42–51).

Model	**FID** (↓)	**KID** (↓)
VAE	142.72 ± 0.64	0.0305 ± 0.0004
DCGAN	47.38 ± 0.65	0.0066 ± 0.0001
DDPM	45.91 ± 0.48	0.0078 ± 0.0002

Table 5 shows the same evaluation protocol on the 20% subset (10,000 images) used during development. On the subset, DDPM and DCGAN performed similarly in FID (45.91 vs. 47.38), with DCGAN leading on KID. After retraining on the full 50,000-image dataset, DCGAN's advantage became substantially larger (FID 30.46 vs. 37.91), indicating that the adversarial objective benefits more from additional data at this resolution. The VAE remained the weakest model in both settings.

6 Conclusion

We implemented and compared three generative model families — VAE, DCGAN, and DDPM — on the ArtBench-10 dataset at 32×32 resolution under a rigorous, reproducible evaluation protocol. The DCGAN achieved the best quantitative results on both FID and KID, benefiting from its adversarial objective and the modest resolution of the task. The DDPM ranked second, offering superior sample diversity at the cost of a slightly higher FID, partly attributable to accelerated DDIM sampling. The VAE ranked last due to the blurriness induced by pixel-wise reconstruction loss.

These results highlight a key trade-off: adversarial models excel at distribution matching at low resolution, while diffusion models provide richer diversity and are expected to scale better to higher resolutions [9]. Future work includes evaluating DDPM with full 1,000-step sampling, training class-conditional models, and extending to 64×64 or higher resolutions.

Use of Generative AI Tools. Generative AI Tools like Claude, Gemini and Chat GPT were used as a code assistant during the development of this project, specifically to help structure training loops, debug PyTorch code, and draft parts of this report. All model implementations, experimental results, and conclusions were verified by the authors.

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